0.1 Ensemble refinement with a generic function $f(\langle g \rangle_P)$

Let's focus on plain ensemble refinement: $\chi^2 + \alpha D_{KL}$: both χ^2 and D_{KL} are convex functions of $P(x)$ with the normalization, so their sum $\mathcal{L}$ is a convex function (sum of convex functions is convex), with just one minimum.

We are interested in minimizing a loss function where the χ^2 is substituted by a generic function of the average values (let's restrict to ensemble refinement, then generalizations can be applied to the refinement of force field and forward models):

$$\mathcal{L}[P] = f(\langle g_i \rangle_P) + \alpha D_{KL}[P|P_0] + \mu(\langle 1 \rangle_P - 1) \tag{1}$$

where we have added the normalization constraint through Lagrange multiplier μ.

By computing the functional derivative of $\mathcal{L}$ with respect to $P(x)$ (including the Lagrange multiplier due to normalization of the weights), we get that the optimal solution has functional form

$$\begin{aligned} \frac{\delta \mathcal{L}[P]}{\delta P(x)} &= \sum_i \frac{\partial f}{\partial \langle g_i \rangle_P} g_i(x) + \alpha \Big(\log \frac{P(x)}{P_0(x)} + 1 \Big) + \mu = 0 \\ \implies P(x) &= \frac{1}{Z_\lambda} P_0(x)\, e^{-\vec{\lambda} \cdot \vec{g}(x)} \end{aligned} \tag{2}$$

with

$$\vec{\lambda} = \frac{1}{\alpha} \frac{\partial f(\langle g_i \rangle_P)}{\partial \langle g_i \rangle_P}. \tag{3}$$

So, as usual we can exploit this parametrization to search for the optimal solution.

Now, in order to get the $\tilde{\Gamma}(\lambda)$ function, which is $\log Z_\lambda$ plus a quadratic term in $\vec{\lambda}$, we have to require $f(\langle g_i \rangle_P)$ to be a quadratic function of the average values. This is because the Legendre transform (which will appear later to get the term in $\tilde{\Gamma}$ which adds to $\log Z_\lambda$) of a non-quadratic function is non-quadratic. So, let $f(\langle g_i \rangle_P)$ be a generic quadratic function:

$$f(\langle g_i \rangle_P) = \frac{1}{2} \sum_{ij} A_{ij} \langle g_i \rangle_P \langle g_j \rangle_P + \sum_i b_i \langle g_i \rangle_P + c \tag{4}$$

with $A_{ij} = A_{ji}$ since quadratic form (otherwise just symmetrize it, since it is the coefficient of a quadratic term). So, it is

$$\alpha\lambda_i = \frac{\partial f}{\partial \langle g_i \rangle_P} = \sum_j A_{ij} \langle g_j \rangle_P + b_i \tag{5}$$

and so the loss function is

$$\begin{aligned}\mathcal{L}(P_\lambda) &= f(\langle g_i \rangle_{P_\lambda}) - \alpha(\log Z_\lambda + \langle \vec{\lambda} \cdot \vec{g}(x) \rangle_{P_\lambda}) \\ &= -\alpha \log Z_\lambda + f(\langle g_i \rangle_{P_\lambda}) - \alpha \vec{\lambda} \cdot \langle \vec{g}(x) \rangle_{P_\lambda} = (*).\end{aligned} \tag{6}$$

Notice that in the second line of the last eq. we have the Legendre transform of $f(\langle g_i \rangle_\lambda)$. By continuing the calculations, we get

$$\begin{aligned}(*) &= -\alpha \log Z_\lambda - \frac{1}{2} \langle g_i \rangle_\lambda A_{ij} \langle g_j \rangle_\lambda + c \\ &= -\alpha \log Z_\lambda - \frac{1}{2} \sum_{ij} (\alpha\lambda_i - b_i) {A^{-1}}_{ij} (\alpha\lambda_j - b_j) + c\end{aligned} \tag{7}$$

which is quadratic in $\vec{\lambda}$ (a part from $\log Z_\lambda$), and so it is the generalization of the $\tilde{\Gamma}$ function to the most general case, still quadratic in the average values and so in $\vec{\lambda}$. One can substitute to A_{ij} and b_j the values they have when $f(\langle g_i \rangle_\lambda) = \frac{1}{2}\chi^2(\langle g_i \rangle_\lambda)$ and go back to exactly the same form of $\tilde{\Gamma}(\lambda)$.

0.2 Mismatch between no. of "observables" and no. of experimental values

Now, the question is what happens when there is a mismatch between the number of "observables" and the number of experimental values.

First case: the number of "observables" N is greater than the number of experimental values M ($M \leq N$); here we mean that the number of observables is always equal to the number of experimental values, but each observable is defined as a linear combination of some basis functions (corresponding to the forward quantities) `forward_qs` whose number is $N \geq M$. In this case, the user-defined function for the forward models can take the input time series for the `forward_qs` and compute the time series of the observables, to be directly compared with the experimental values; by doing so, there is a dimensionality reduction coded by the forward model functions written by the user (from N time series to M).

Second case: the number of "observables" N is lower than the number of experimental values M ($M \geq N$); case limit: two experimental values are imposed for the same observable. In this case, we can reduce the number of $\vec{\lambda}$ components to the number of experimental values and then go back to eq. 7 to get a quadratic function in $\vec{\lambda}$. This can be implemented as a Python function to be applied when the n. of experimental values is greater than the number of basis functions from which the observables are computed through linear combination (the linear combination is important, because the case is different when imposing two values for sine and cosine of the same time series for the angles). We have

$$\begin{aligned} \sum_i \Big(\sum_j M_{ij} \langle g_j \rangle_\lambda - g_{i,exp} \Big)^2 &= \sum_{j,j'} \langle g_j \rangle_\lambda \sum_i M_{ij} M_{ij'} \langle g_{j'} \rangle_\lambda \\ - \sum_j \langle g_j \rangle_\lambda \cdot 2 \sum_i M_{ij} g_{i,exp} + \sum_i g_{i,exp}^2 \end{aligned} \tag{8}$$

and so the identification with the generic quadratic form in eq. 4 is made by

$$A_{jj'} = \sum_i M_{ij} M_{ij'} \quad b_j = -2 \sum_i M_{ij} g_{i,exp} \quad c = \sum_i g_{i,exp}^2. \tag{9}$$

For example, look at the limiting case of two experimental values constrained for the same observable. By doing this manipulation of the χ^2, we get a quadratic form in $\langle g \rangle$ (one variable) and so just one component of $\vec{\lambda}$.

0.3 Alternative derivation of the $\tilde{\Gamma}$ function

In this section, we derive the $\tilde{\Gamma}$ function with explicit usage of the Legendre transform of the $\chi^2[P]$ and of the Gibbs variational relation

$$\min_P (\langle h(x) \rangle_P - S[P||P_0]) = -\log \langle e^{-h(x)} \rangle_{P_0} \tag{10}$$

which has a deep thermodynamic interpretation when substituting to $h(x)$ the energy $\beta E(x)$ (the minimum value of the free energy $\langle \beta E(x) \rangle_P - S[P||P_0]$ is actually the log partition function – with a minus?). This deep relation is quite easy to proof, indeed:

$$\begin{aligned} &\frac{\delta}{\delta P(x)} [\langle h(x) \rangle_P + D_{KL}[P||P_0] + \mu(\langle 1 \rangle_P - 1)] \\ &= h(x) + \log \frac{P(x)}{P_0(x)} + 1 + \mu = 0 \end{aligned} \tag{11}$$

and so $P^*(x) \propto P_0(x)\, e^{-h(x)}$ and

$$(\langle h(x)\rangle_P + D_{KL}[P||P_0])|_{P^*} = -\log Z_{P^*} = -\log\langle e^{-h(x)}\rangle_{P_0}. \tag{12}$$

A bit more difficult is to proof that it is actually the minimum value and not an arbitrary stationary point.

This will allow us to derive the $\tilde{\Gamma}$ function without relying on the functional derivatives of $\mathcal{L}$, in a way that appears more easy to understand (since the $\vec{\lambda}$ coefficients appear directly in a minimization problem).

So, firstly let's apply the Legendre transform to the χ^2: this means we can rewrite the χ^2 as a minimization problem over the $\vec{\lambda}$ coefficients, allowing to get a linear relation for $\langle g\rangle_P$ rather than quadratic. It is:

$$\begin{aligned}\mathcal{L}(P) &= \frac{1}{2}\sum_i\Big(\frac{\langle g_i(x)\rangle_P - g_{i,exp}}{\sigma_{i,exp}}\Big)^2 + \alpha D_{KL}[P||P_0] \\ &= \max_{\vec{\lambda}}\Big[\vec{\lambda}\cdot(\langle\vec{g}(x)\rangle_P - \vec{g}_{exp}) - \frac{1}{2}\sum_i \sigma_{i,exp}^2\lambda_i^2\Big] + \alpha D_{KL}[P||P_0]\end{aligned} \tag{13}$$

(easy to proof relying on quadratic equations). Now, since D_{KL} does not depend on $\vec{\lambda}$, we can put it into the $\max_{\vec{\lambda}}$ and then take the $\min_P \mathcal{L}(P)$:

$$\min_P \mathcal{L}(P) = \min_P \max_{\vec{\lambda}}\Big[\vec{\lambda}\cdot(\langle\vec{g}(x)\rangle_P - \vec{g}_{exp}) - \frac{1}{2}\sum_i \sigma_{i,exp}^2 + \alpha D_{KL}[P||P_0]\Big]. \tag{14}$$

Now, the tricky point: by virtue of the Von Neumann minimax theorem, since this function $f(\lambda, P)$ is concave in $\vec{\lambda}$ and convex in $P(x)$ (proved), we can exchange $\max_{\vec{\lambda}}$ with $\min_P$ and then make use of the Gibbs variational relation

$$\begin{aligned}\min_P \mathcal{L}(P) &= \max_{\vec{\lambda}}\min_P\Big[\vec{\lambda}\cdot(\langle\vec{g}(x)\rangle_P - \vec{g}_{exp}) - \frac{1}{2}\sum_i \sigma_i^2\lambda_i^2 + \alpha D_{KL}[P||P_0]\Big] \\ &= \max_{\vec{\lambda}}\Big[-\alpha\log\langle e^{-\vec{\lambda}/\alpha\cdot\vec{g}(x)}\rangle_{P_0} - \vec{\lambda}\cdot\vec{g}_{exp} - \frac{1}{2}\sum_i\sigma_i^2\lambda_i^2\Big] = -\alpha\min_{\vec{\lambda}}\tilde{\Gamma}(\vec{\lambda})\end{aligned} \tag{15}$$

where in the last passage we substituted $\vec{\lambda} \to \vec{\lambda}/\alpha$.

However, with this derivation we cannot prove that the optimal solution can be parametrized by $\vec{\lambda}$ coefficients independently on the functional form of the term $f(\langle g(x)\rangle_P)$ summed to the relative entropy.

0.4 Application to the case of pH refinement

Starting point: constant pH MD simulations. A MD simulation with constant pH samples the grand canonical distribution

$$P_{GC}(x) = \frac{1}{Z_{GC}} e^{-\beta(E(x)-\mu N(x))} \tag{16}$$

so the probability to be in a subtrajectory (assumed ergodic sampling) with given protonation state j (i.e., fixed $N(x) = N_j$) is

$$P(j) = \frac{1}{Z_{GC}} \int_{x\in j} dx\, e^{-\beta(E(x)-\mu N_j)} = \frac{e^{-\beta F_j}}{Z_{GC}} e^{\beta\mu N_j} \tag{17}$$

and so the ratio between the probabilities to be in a protonation state j and j' will depend also on the canonical free-energy of the two protonation states, that has therefore to be estimated before moving on.

In this case, we are interested in minimizing this loss function

$$\begin{aligned}\mathcal{L}(\{P_j\}, \vec{\pi}) =& \frac{1}{2}\sum_{k=1}^{N_{obs}} \sum_{i\in\sigma(k)} \Big(\frac{\sum_j w_{ij}\langle g_k\rangle_{P_j} - g_{ki,exp}}{\sigma_{ki,exp}}\Big)^2 \\ &+ \sum_j \alpha_j D_{KL}[P_j||P_{0j}] + \alpha D_{KL}[\vec{\pi}|\vec{\pi}_0]\end{aligned} \tag{18}$$

with normalization of $P_j(x)$, $\vec{\pi}$ and w_{ij} (the relation between w_{ij} and π_j is given by the grand-canonical ensemble; also α_j might depend on $\vec{\pi}$, for example as $\alpha_j = \alpha\pi_j$, but perhaps it has more physical meaning to keep all them $\alpha_j = \alpha$); $\sigma(k)$ are the pH indices for which the k-th observable has been measured.

Minimizing this loss function over the ensembles P_j is equivalent to maximize the following loss function (same spirit of $\tilde{\Gamma}$) over the $\vec{\lambda}$ coefficients (one for each g_{exp} value, independently on the number of protonation states $\#\{P_j\}$):

$$\begin{aligned}\tilde{\mathcal{L}}(\vec{\lambda}, \vec{\pi}) = &- \sum_k \sum_{i\in\sigma(k)} \Big(\frac{1}{2}\sigma^2_{ki,exp}\lambda^2_{ki} + \lambda_{ki} g_{ki,exp}\Big) - \sum_j \log Z_j(\vec{\lambda}) \\ &+ \alpha D_{KL}[\vec{\pi}||\vec{\pi}_0]\end{aligned} \tag{19}$$

where the ensemble P_j is parametrized by λ as

$$P_j(x;\vec{\lambda}) = \frac{1}{Z_j(\vec{\lambda})} P_{0j}(x) \exp\Big\{-\frac{1}{\alpha_j}\sum_k \sum_{i\in\sigma(k)} \lambda_{ki} w_{ij} g_k(x)\Big\}. \tag{20}$$

Notice the $\vec{\lambda}$ coefficients are present according with the experimental values, not with the structural (protonation) ensembles P_j.

This simplifies to our usual case, in which we have experimental values corresponding to same pH value (but different molecular systems).

So, in the end we have to solve a minimax problem $\min_{\vec{\pi}} \max_{\vec{\lambda}}$; contrary to the case of combination with force-field refinement, here the function of $\vec{\pi}$ is convex and so we can exchange min/max and the possibility for an internal minimization over $\vec{\pi}$ and external over $\vec{\lambda}$ is allowed (?). However, we retain it to be less efficient than an internal minimization over $\vec{\lambda}$. This in general will depend on the number of components of the two vectors $\vec{\lambda}$ and $\vec{\pi}$.

Relation between w_{ij} and $\vec{\pi}$

$$P(j) = \frac{e^{-\beta F_j}}{Z_{GC}}\, e^{\beta \mu N_j} \tag{21}$$

$$P(j)|_{\mu=\mu_0} = \pi_j \propto e^{-\beta F_j}\, e^{\beta\mu_0 N_j} \tag{22}$$

so it is

$$P(j)|_{\mu_i} = w_{ij} \propto e^{-\beta F_j}\, e^{\beta\mu_i N_j} \propto \pi_j\, e^{\beta\Delta\mu_i N_j} \tag{23}$$

with $e^{-\beta F_j} = Z_j$ partition function for the j-th canonical ensemble.

Derivation of P_j and $\tilde{\mathcal{L}}$

The derivative of $\mathcal{L}$ in eq. 18 with respect to $P_j(x)$ is:

$$\frac{\delta\mathcal{L}(\{P_j\},\vec{\pi})}{\delta P_j(x)} = \sum_k \sum_{i\in\sigma(k)} \frac{\sum_{j'} w_{ij'}\langle g_k\rangle_{P_{j'}} - g_{ki,exp}}{\sigma^2_{ki,exp}}\, w_{ij}\, g_k(x) + \alpha_j\Big(\log\frac{P_j(x)}{P_{0j}(x)} + 1\Big) + \mu_j \tag{24}$$

where μ_j is the Lagrange multiplier for the normalization of $P_j(x)$. From this, we derive that the optimal solution $P_j(x)$ takes the functional form described in Eq. 20 where the set of coefficients λ_j for the optimal solution is determined by the set of implicit equations

$$\lambda_{ki} = \frac{\sum_j w_{ij}\langle g_k(x)\rangle_{P_j} - g_{ki,exp}}{\sigma^2_{ki,exp}} \tag{25}$$

Rather than solving this set of implicit equations, as usual we minimize the loss function taking advantage of this parametrization. Substituting this functional form in $\mathcal{L}(\{P_j\}, \vec{\pi})$ we get $\tilde{\Gamma}$ as a maximization problem (it can be proved also as in the previous section, by applying Legendre transform to the χ^2 so that the maximization problem is soon evident).

Critical issue – solved

The weights $\vec{\pi}$ are not independent on the reweighting of the canonical distributions, since they depend on the partition functions Z_j. Can I perform this nested minimization anyway?

For example, if we consider a correction to the energy parametrized by some coefficients $\vec{\phi}$, then the loss function is a single minimization over $\vec{\phi}$, where all the three terms $(\chi^2, D_{KL}[P_j||P_{0j}], D_{KL}[\vec{\pi}||\vec{\pi}_0])$ depend on ϕ.

$$\mathcal{L}(\phi) = \frac{1}{2}\chi^2(\phi) + \sum_j \alpha_j D_{KL}[P_j(\phi)||P_{0j}] + \alpha D_{KL}[\vec{\pi}(\phi)||\vec{\pi}_0] \tag{26}$$

So, we do not have an internal minimization over the set of ensembles (described in this case by ϕ) and an external one over $\vec{\pi}$, since the latter depends on ϕ. Now, the question is: what happens without a parametrization for our set of ensembles? Are we still able to get a useful dimensionality reduction in terms of the $\vec{\lambda}$ coefficients?

Yes: indeed, even if the loss function depends on the energy corrections at each frame in each canonical ensemble corresponding to a protonation state, we can split this dependence into *local* and *global* contributions. The *local* contributions preserve the free energy differences between different protonation states (namely, w_{ij}), while the *global* ones do not. The global contributions correspond to a global shift of the energy at each protonation state (this shift can be different for each canonical ensemble), and so they change the partition functions Z_j. The canonical ensembles P_j depends only on the local contributions (we can fix $Z_j = 1$) whereas the w_{ij} only on the global contributions. Hence, the $D_{KL}[P_j||P_{0j}]$ terms depend only on the local contributions (they do not change upon different global contributions, since they focus on each canonical ensemble and not on comparing them with each other). The χ^2 term depends both on the local contributions (through $\langle g(x)\rangle_P$) and the global contributions (through w_{ij}). The $D_{KL}[\pi||\pi_0]$ depends only on the global contributions. Hence:

$$
\begin{aligned}
\mathcal{L}(\{\delta E_j(x)\}) =& \frac{1}{2}\chi^2(\{\delta E_j(x)\}) + \sum_j \alpha_j D_{KL}[P_j(\{\delta E_j(x)\}) \,||\, P_{0j}(x)] \\
&+ \alpha D_{KL}[\vec{\pi}(\{\delta E_j(x)\}) \,||\, \vec{\pi}_0]
\end{aligned}
\tag{27}
$$

and we can minimize over $\{\delta E_j(x)\}$ by splitting into local and global contributions:

$$
\min_{\{\delta E_j(x)\}} \mathcal{L}(\{\delta E_j(x)\}) = \min_{\text{global}} \min_{\text{local}} \mathcal{L}(\{\delta E_j(x)\}) = \min_{\vec{\pi}} \min_{\{P_j(x)\}} \mathcal{L}(\vec{\pi}, \{P_j(x)\})
\tag{28}
$$

Apply this reasoning to the ensemble refinement for $\Delta\Delta G$ (or ΔG) data, we have

$$
\mathcal{L}(\{\delta E_j(x)\}) = \frac{1}{2}\chi^2(\{Z_j\}) + R(\{P_j\})
\tag{29}
$$

where the χ^2 term depends only on global contributions (shift of the relative free energies) and the regularization R (D_{KL} over P_j) depends only on local contributions. So, we have a separation of local and global contributions (while in the case of pH refinement the χ^2 couples local and global contributions). Consequently, the optimal solution corresponds to zero local corrections, just global ones:

$$
\begin{aligned}
&\min_{\{\delta E_j(x)\}} \mathcal{L}(\{\delta E_j(x)\}) = \min_{\{Z_j\}} \min_{\{P_j\}} \Big[\frac{1}{2}\chi^2(\{Z_j\}) + R(\{P_j\})\Big] \\
&= \min_{\{Z_j\}} \frac{1}{2}\chi^2(\{Z_j\}) + \min_{\{P_j\}} R(\{P_j\}) = \min_{\{Z_j\}} \frac{1}{2}\chi^2(\{Z_j\}) + 0.
\end{aligned}
\tag{30}
$$

0.5 Inclusion of experimental knowledge about relative populations

Beyond ensemble refinement and force-field fitting based on comparison with experimental values corresponding to ensemble averages (over canonical distributions), we include experimental knowledge about free-energy differences, or equivalently relative populations, as in the case of denaturation experiments (measuring $\Delta\Delta G$) or quantities depending on the pH (grand-canonical distribution). In the most general case, the loss function is

$$
\mathcal{L}(\{P, Z\}) = \frac{1}{2}\chi^2[\{P, Z\}] + \alpha_1 R[\{P\}] + \alpha_2 R[\{Z\}]
\tag{31}
$$

where $\{P\}$ is the set of canonical ensembles (one for each physical states) like a protonation state or methylated/wild-type single-strand/duplex) and $\{Z\}$ the set of corresponding partition functions, related to population ratios (or equivalently free-energy differences) between different states, each corresponding to a canonical ensemble P_i. Using different variables, this loss function can be expressed in terms of energy shifts $\{P(x), Z\} = \{\delta E(x)\} = \{\delta E(x), \Delta E\}$ where the number of variables is given by the number of frames x ($P(x)$ has one less due to normalization, and we take a global shift ΔE reducing the number of free $\delta E(x)$.

In the case of pH-dependent data, the χ^2 couples P and Z quantities, while in the case of plain ensemble refinement the χ^2 does not depend on Z (namely to global energy shifts – P are normalized to 1) and in the case of denaturation experiments χ^2 depends only on Z. Let's examine these different scenarios one by one, either for ensemble refinement or force-field fitting.

- Ensemble refinement for pH-dependent data: in this case, P_i and Z_i are coupled by the χ^2 and the loss function is the same as in previous equation, so its minimization is

$$\min_{P,Z} \mathcal{L}(P, Z) = \min_{Z} \min_{P} \mathcal{L}(P, Z) \tag{32}$$

  so we can perform an inner minimization over P (that leads to the $\vec{\lambda}$ parametrization as described above) and an external one over Z (the latter corresponds to minimizing over the relative populations).

- Ensemble refinement for ensemble averages: in this case the χ^2 does not depend on global energy shifts $\chi^2 = \chi^2(P)$, so we can neglect the regularization on Z (keep $Z = Z_0$, but they are gauge invariant quantities) and so we have to minimize the loss function only on P, getting the $\vec{\lambda}$ parametrization.

- Ensemble refinement for relative populations: here, $\chi^2 = \chi^2(Z)$ as in the case of denaturation experiments, so we can split the minimization

$$\min_{P,Z} \mathcal{L}(P, Z) = \min_{Z} \Big[\frac{1}{2}\chi^2(Z) + \alpha_2 R(Z)\Big] + \min_{P} \alpha_1 R(P) \tag{33}$$

  which means $P = P_0$, namely optimize global energy shifts only, loosing any information about ensemble refinement, since it focuses only on relative populations Z, so we have to minimize only over Z. This case is specular to the previous one. If I decide to fix global energy shifts, the $\chi^2[Z]$ is constant, so no improvement; however, with force-field fitting, the global energy shifts can be either left completely free to vary or constrained to be parametrized by ϕ (for example, if the grand-canonical distribution is parametrized by ϕ, we can decide to include or not global energy shifts, one for each canonical ensemble).

- Ensemble refinement with $\chi^2(P, Z) = \chi^2(P) + \chi^2(Z)$ (in this case we have two separate minimizations)

- Force-field fitting: here we have two different cases, depending on whether we allow for independent global energy shifts or not. If the Z and P are not coupled by the χ^2 the loss function is

$$\mathcal{L} = \frac{1}{2}\chi^2[Z] + \frac{1}{2}\chi^2[P(\phi)] + \alpha_1 R[P(\phi)] + \alpha_2 R[Z] \tag{34}$$

  and we have two different cases: if global energy shifts are allowed to be independent on the force-field correction ϕ, the Z variables are independent from ϕ and we have two separate minimizations (that reduce to a single one if χ^2 depends only on Z or on P), while if global energy shifts are completely determined by ϕ there is a single minimization over ϕ.

- Force-field fitting with Z and P coupled by the χ^2: if Z are free to vary independently on ϕ, we have to minimize both over Z and ϕ, for example we can minimize internally over ϕ and externally over Z, or vice versa. On the contrary, if Z are determined by ϕ, there is a single minimization (as in the case of alchemical calculations done with Valerio).